

ADDITIONAL PROBLEM SHEET 6

1

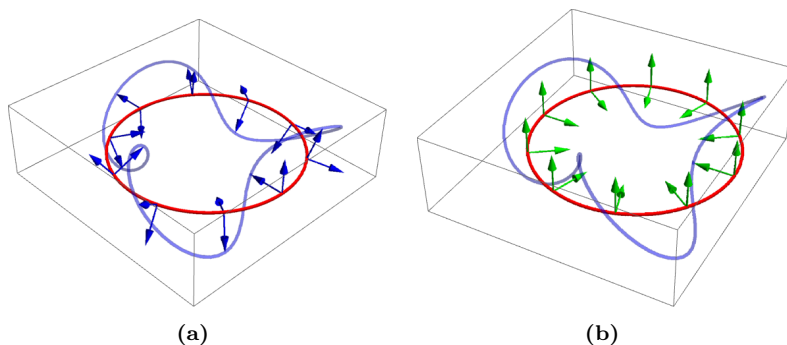


Figure 1: Depictions of a possible minicircle DNA model. (a) The red circle is the tube's axis. The blue curve represents the tip of the vector $\mathbf{d}_1$ and indicates the extent to which the tube is twisted. (b) the green vectors represent the twist free basis $(\mathbf{E}_1, \mathbf{E}_2)$.

Consider a simple model for a DNA mini-circle as a twisted elastic tube. The tube's axis is assumed to be a circle

$$\mathbf{r}(t) = (R \cos(2\pi t), R \sin(2\pi t), 0), t \in [0, 1] \quad (1)$$

where R is the radius of the circle. We assume tube can be modelled as an unstretchable tube satisfying the following kinematic equations

$$\frac{d}{ds} \begin{pmatrix} \mathbf{r} \\ \mathbf{d}_1 \\ \mathbf{d}_2 \\ \mathbf{d}_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & u_3 & -u_2 \\ 0 & -u_3 & 0 & u_1 \\ 0 & u_2 & -u_1 & 0 \end{pmatrix} \begin{pmatrix} \mathbf{r} \\ \mathbf{d}_1 \\ \mathbf{d}_2 \\ \mathbf{d}_3 \end{pmatrix}. \quad (2)$$

and the following equilibrium equations

$$\begin{aligned} \frac{d}{ds} \mathbf{n} &= \mathbf{0}, \\ \frac{d}{ds} \mathbf{m} + \mathbf{d}_3 \times \mathbf{n} &= \mathbf{0}. \end{aligned} \quad (3)$$

where $\mathbf{n}$ is the internal force of the tube, $\mathbf{m}$ the internal couple. We assume the following constitutive laws for the system

$$\mathbf{m} = Au_1 \mathbf{d}_1 + Bu_2 \mathbf{d}_2 + Cu_3 \mathbf{d}_3, \quad \mathbf{n} = n_1 \mathbf{d}_1 + n_2 \mathbf{d}_2 + n_3 \mathbf{d}_3. \quad (4)$$

(a) Find the arclength parametrization $\mathbf{r}(s)$ of this curve.

(b) Express the $\mathbf{d}_i$ components of the equations (??).

(c) It can be shown that the kinematic parameters can be written as

$$u_1 = \frac{\sin(\phi(s))}{R}, \quad u_2 = \frac{\cos(\phi(s))}{R}, \quad u_3 = \frac{d\phi}{ds}. \quad (5)$$

If $\phi(s) = 0, \forall s$ then the pair $(\mathbf{d}_1, \mathbf{d}_2)$ are parallel transported around $\mathbf{r}(s)$, as shown in Figure ??(b), this represents an untwisted tube. An example of a non zero twisted tube (basis pair) is shown in Figure ??(a). State the (two) boundary conditions on ϕ on the assumption that the deformation of the tube is differentiable.

(d) Demonstrate that the twist function $d\phi/ds$ takes the following implicit form:

$$\frac{d\phi}{ds} = \pm \sqrt{2 \left(D + \frac{\cos(2\phi)}{4R^2} (B - A) \right) / C}, \quad (6)$$

where D is a constant of integration.

(e) Find a condition on the constants A and B for the possibility of solutions to the full system of equations when there is non-zero twisting. You may assume that the O.D.E for ϕ obtained in the previous question has solutions. (Hint: you should not need to solve the equations).

2

The kinematics of the rod are described by the set of vectors $\mathbf{r}(s), \mathbf{d}_i(s) [0, L] \rightarrow \mathbb{R}^3, i = 1, 2, 3$ and the following system of ordinary differential equations

$$\frac{d}{ds} \begin{pmatrix} \mathbf{r} \\ \mathbf{d}_1 \\ \mathbf{d}_2 \\ \mathbf{d}_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & \nu \\ 0 & 0 & u_3 & -u_2 \\ 0 & -u_3 & 0 & u_1 \\ 0 & u_2 & -u_1 & 0 \end{pmatrix} \begin{pmatrix} \mathbf{r} \\ \mathbf{d}_1 \\ \mathbf{d}_2 \\ \mathbf{d}_3 \end{pmatrix},$$

with the u_i, ν smooth scalar functions of s representing the system's curvatures and stretching

The equations of equilibrium of this body are given by the following vector equations:

$$\begin{aligned} \frac{d}{ds} \mathbf{n} &= \mathbf{0}, \\ \frac{d}{ds} \mathbf{m} + \frac{d\mathbf{r}}{ds} \times \mathbf{n} &= \mathbf{0}. \end{aligned} \quad (7)$$

The following choice of constitutive law for the internal couple is assumed

$$\mathbf{m} = Au_1 \mathbf{d}_1 + Au_2 \mathbf{d}_2 + Cu_3 \mathbf{d}_3,$$

with $A, C > 0$ constants. To complete the system (??) we assume the following constitutive law relating the axial force and the stretching:

$$n_3(s) = E(\nu(s) - 1). \quad (8)$$

where E is the Young's modulus of the elastic rod.

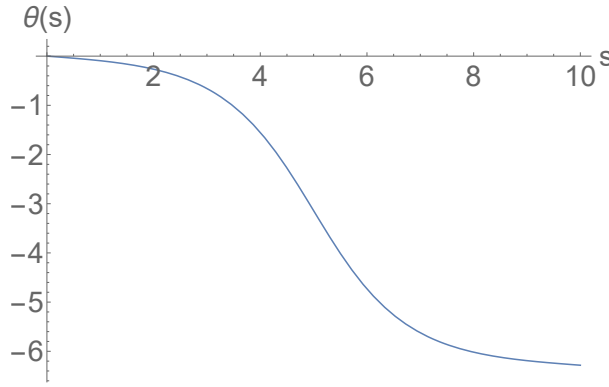


Figure 2: A function $\theta(s)$, $s \in [0, 10]$ which is a solution to the tube system considered in Question 5

- (a) Write out the system (??) in component form in the $(\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3)$ basis.
- (b) Now assume that $u_2 = u_3 = 0$, how does this constraint the tube's geometry?
- (c) Use the assumption that $u_2 = u_3 = 0$ to reduce the system (??) to a pair of non-linear ordinary differential equations for ν and u_1 .
- (d) Verify that a straight vertical rod with arbitrary *constant compression* is in equilibrium for this model and given an expression for the compression under an axial load N applied to the rod.
- (e) Demonstrate the rod is unstable to the forces with the following values for integer n :

$$N = \frac{1}{2} \left(E \pm \sqrt{E} \sqrt{E - \frac{4\pi^2 A n^2}{L^2}} \right) \quad (9)$$

To do this assume the rod remains planar under small perturbations about this equilibrium (Hint: perform a linear stability analysis).

3

Consider a slender elastic rod whose central axis is represented by a curve $\mathbf{r}(s) : [0, L] \rightarrow \mathbb{R}^3$, where s is the body's arclength. The coordinates of $\mathbb{R}^3$ take the form (x, y, z) with unit vectors $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$ respectively. A force $\mathbf{n}(s)$ represents the internal force acting on each cross-section of the the body and $\mathbf{m}(s)$ represents the internal couple acting on each cross section. Net forces and couples can be applied at $s = 0$ and $s = L$.

The kinematics of the rod are described by the set of vectors $\mathbf{r}(s), \mathbf{d}_i(s) : [0, L] \rightarrow \mathbb{R}^3, i = 1, 2, 3$ and the following system of ordinary differential equations

$$\frac{d}{ds} \begin{pmatrix} \mathbf{r} \\ \mathbf{d}_1 \\ \mathbf{d}_2 \\ \mathbf{d}_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & u_3 & -u_2 \\ 0 & -u_3 & 0 & u_1 \\ 0 & u_2 & -u_1 & 0 \end{pmatrix} \begin{pmatrix} \mathbf{r} \\ \mathbf{d}_1 \\ \mathbf{d}_2 \\ \mathbf{d}_3 \end{pmatrix},$$

with the u_i smooth scalar functions of s representing the system's curvatures.

The equations of equilibrium of this body are given by the following vector equations:

$$\begin{aligned}\frac{d}{ds}\mathbf{n} &= \mathbf{0}, \\ \frac{d}{ds}\mathbf{m} + \mathbf{d}_3 \times \mathbf{n} &= \mathbf{0}.\end{aligned}\tag{10}$$

The following choice of constitutive law for the internal couple is assumed

$$\mathbf{m} = Au_1\mathbf{d}_1 + Au_2\mathbf{d}_2 + Cu_3\mathbf{d}_3,$$

with $A, C > 0$ constants. Further assume the rod is a planar curve and the pair $\mathbf{d}_3$ and $\mathbf{d}_1$ are parameterised by a function $\theta(s)$:

$$\mathbf{d}_3(s) = (\cos \theta(s), 0, \sin \theta(s)), \quad \mathbf{d}_1(s) = (-\sin \theta(s), 0, \cos \theta(s)).$$

Assume the rod is never twisted ($u_3 = 0$). Finally, consider the rod to be attached to a wall at $s = 0$ at an angle $\theta = 0$ and that a load $N\hat{\mathbf{x}}$ pushing on the rod is applied at $s = L$.

- (a) Show that the system (??) can be reduced to the following non-linear ODE for θ :

$$A\frac{d^2\theta}{ds^2} - N\sin(\theta) = 0.\tag{11}$$

- (b) Using the following integral identity

$$\int^\theta \frac{d\theta}{\sqrt{C_1 - \gamma \cos \theta}} = \frac{2}{\sqrt{C_1 - \gamma}} F\left(\frac{\theta}{2} \middle| \frac{2\gamma}{C_1 - \gamma}\right),$$

where

$$F(x|k) = \int_0^x \frac{dx'}{\sqrt{1 - k^2 \sin^2 x'}}$$

is the elliptic integral of the first kind, integrate (??) twice in order to find an implicit relationship between θ and s and a constant of integration C_1 . In doing so you should impose the stated boundary condition on θ at $s = 0$.

- (c) State a constraint on the value of C_1 in terms of the applied load N and bending stiffness A .
- (d) The elliptic integral of the first kind has an inverse, the so-called Jacobi amplitude function $\text{am}(x|k)$ defined by

$$F(\text{am}(x|k)|k) = x.$$

Use this relation to obtain an explicit solution for θ .

- (e) An example solution for $\theta(s)$, $s \in [0, 10]$ is shown in Figure ?? . Describe/sketch the tube configuration $\mathbf{r}(s)$, $s \in [0, 10]$ corresponding to this solution. Is it physically realistic?
- (f) Consider a similar model except with the bending coefficient $A(s)$ dependent on the arclength. What does this represent physically and how would it alter the equation obtained in (a) (assuming all else stays equal)?